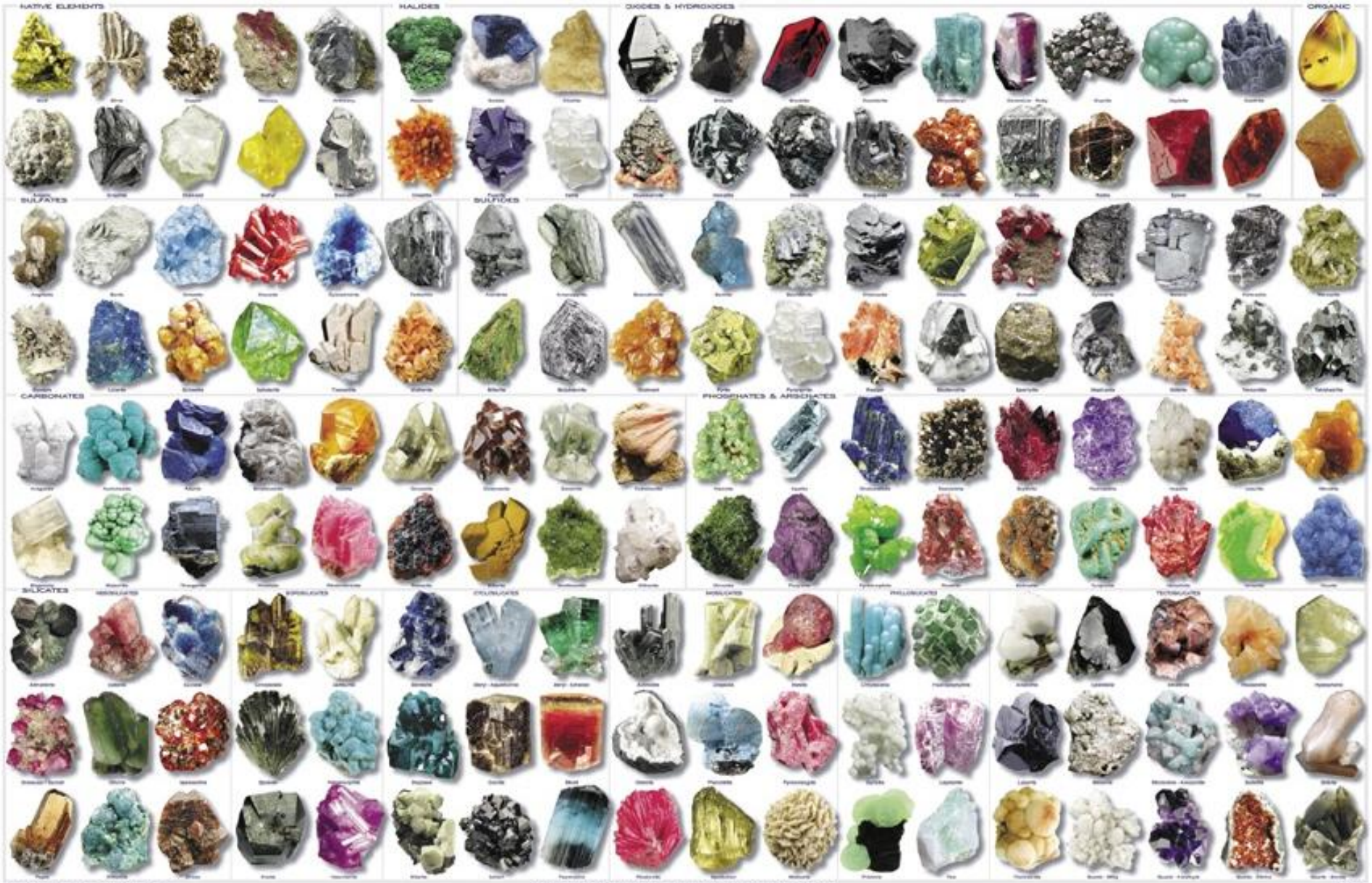
Minerals


NATIVE ELEMENTS
HALIDES
OXIDES & HYDROXIDES
ORGANIC
SULFATES
SULFIDES
CARBONATES
PHOSPHATES & ARSENATES
SILICATES

Minerals: Building blocks of rocks

- **Definition of a mineral:**
  - **Naturally occurring**
  - **Inorganic solid**
  - **Ordered internal molecular structure**
  - **Definite chemical composition**
- **Definition of a rock:**
  - **A solid aggregate or mass of minerals**

How do minerals form

- Crystal is any natural solid with an ordered, repetitive, atomic structure.

- Minerals form through crystallization. The molecules, or ions, chemically bond to form an orderly structure.

- Example:

Dissolved salt in water solution

Evaporation →

Formation of salt crystals

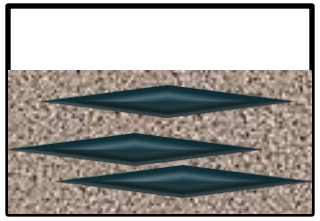
Hot magma

Temperature drop →

Formation of quartz crystals

Mudstone

Temperature and Pressure change →



Formation of mica crystals

Mineral properties

Hardness: - How hard a mineral is depends primarily on its crystal structure.
- Example: Diamond & graphite
- Mohs hardness scale

Habit/ Shape:
-Characteristic shape
-Determined by crystal structure
- Types: Bladed
Platy
Prismatic
Banded
Botryoidal

Cleavage

Optical properties:
- Colour
- Streak
- Luster

TABLE 2.2 **Mohs Scale of Hardness**

Relative Scale		Mineral	Hardness of Some Common Objects
Hardest	10	Diamond	
	9	Corundum	
	8	Topaz	
	7	Quartz	
	6	Potassium Feldspar	
	5	Apatite	5.5 Glass, Pocketknife
	4	Fluorite	
	3	Calcite	3 Copper Penny
	2	Gypsum	2.5 Fingernail
Softest	1	Talc	

Mineral properties: Shape

Bladed kyanite

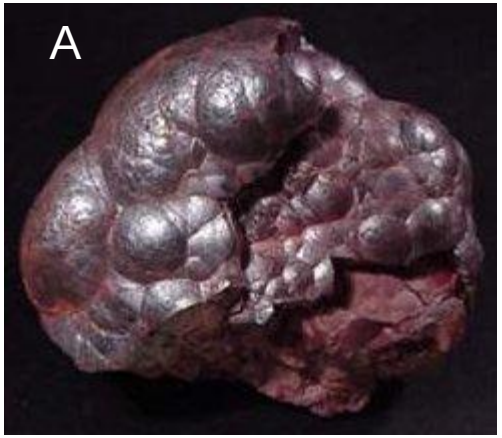
Botryoidal hematite

Prismatic quartz

Banded agate

Fibrous okenite

Identify the following shapes:

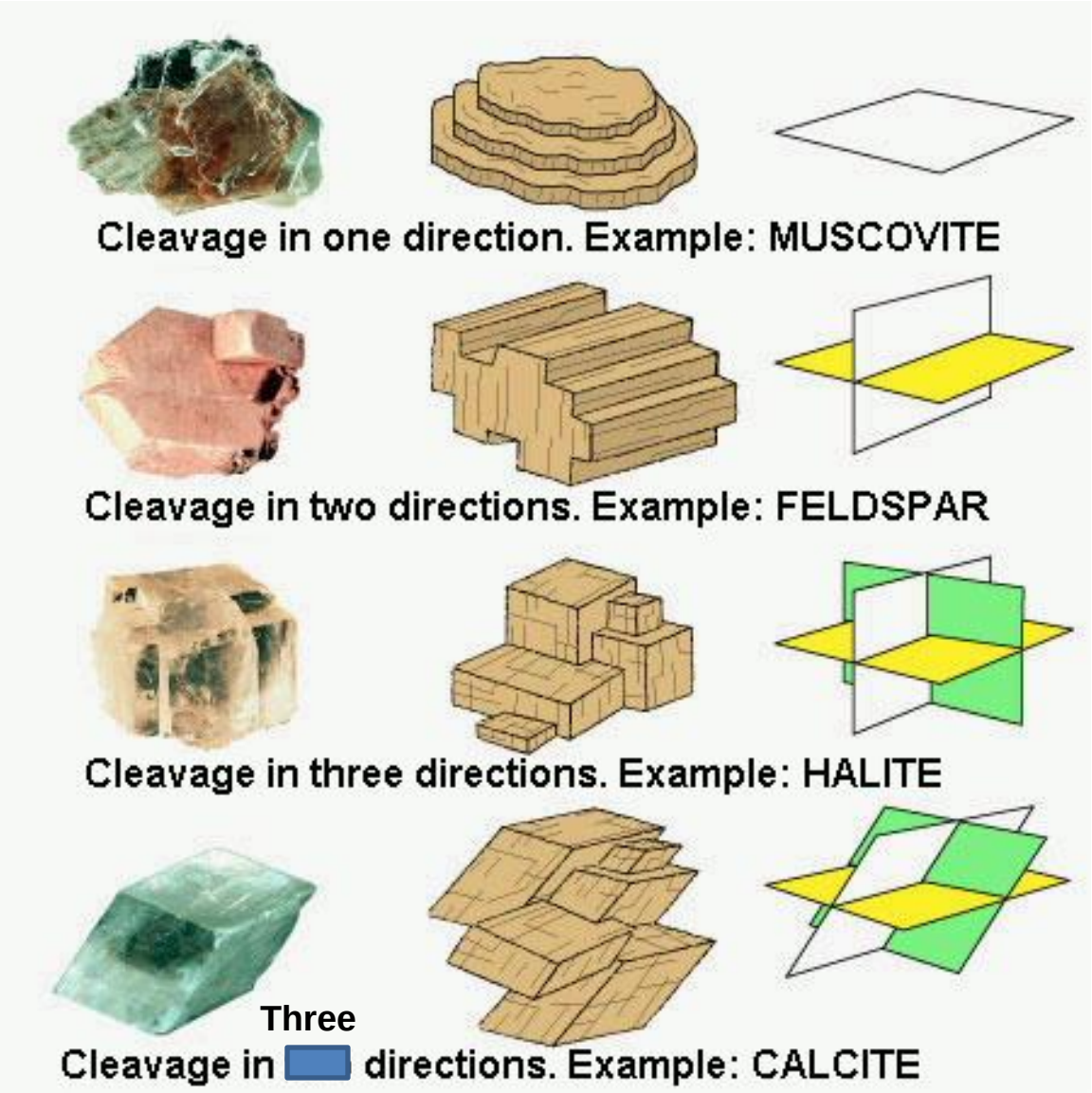


Mineral properties: cleavage

- Preferred plane of breakage.

- Creates a smooth plane.

- Generated by atomic arrangements.



Mineral properties

Optical property:

Varieties of quartz

Color: Rarely a diagnostic property.

Streak: -The color of powdered mineral.
- Diagnostic property.
- Rub the mineral across unglazed porcelain.

Red-brown streak of the mineral hematite.

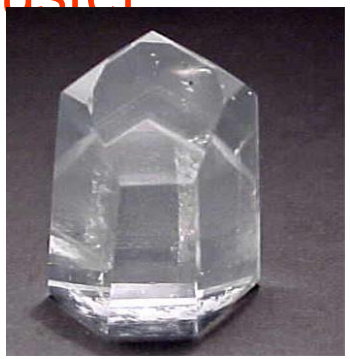
Luster: - The appearance of light reflected from mineral surface.

Types: 1. Metallic

2. Non-metallic : Dull/Earthy
Glassy
Pearly
Silky

Different types of luster

Metallic luster of pyrite



Glassy luster of quartz

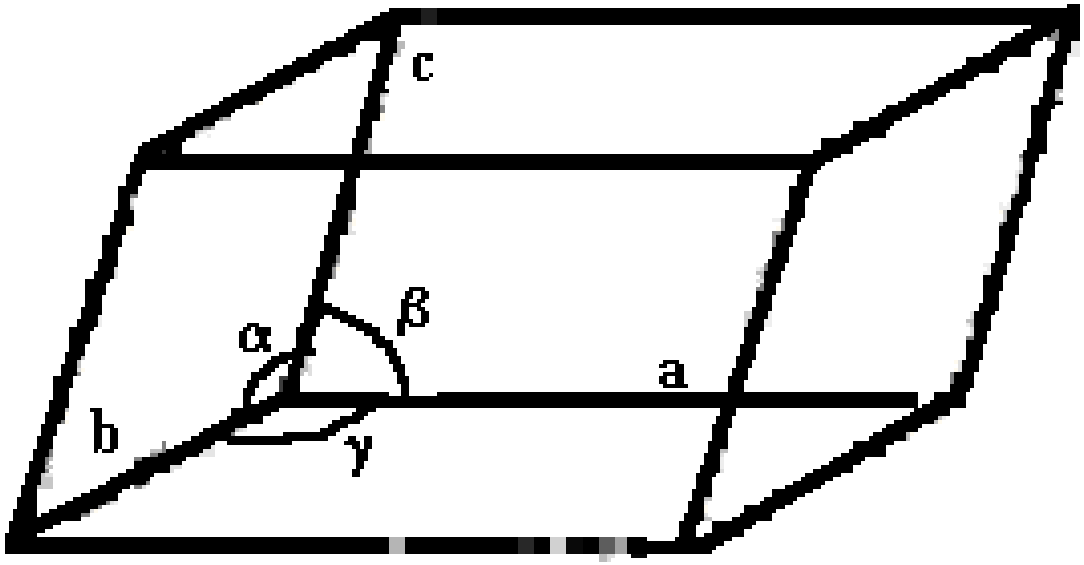
Earthy luster of limonite

Pearly luster of talc

Silky luster of siderite

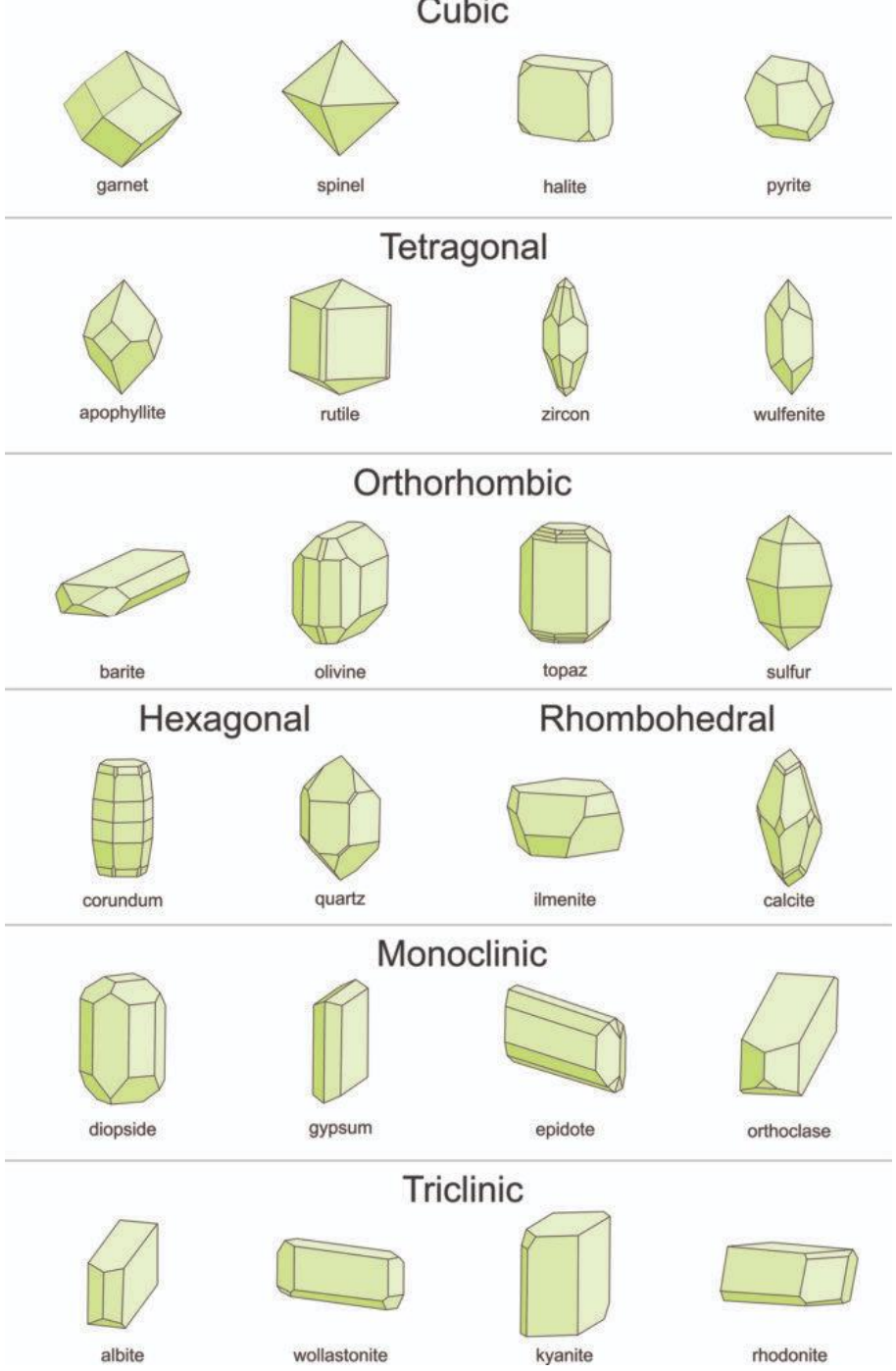
Crystals are made of infinite number of unit cells

Unit cell is the smallest unit of a crystal, which, if repeated, could generate the whole crystal.

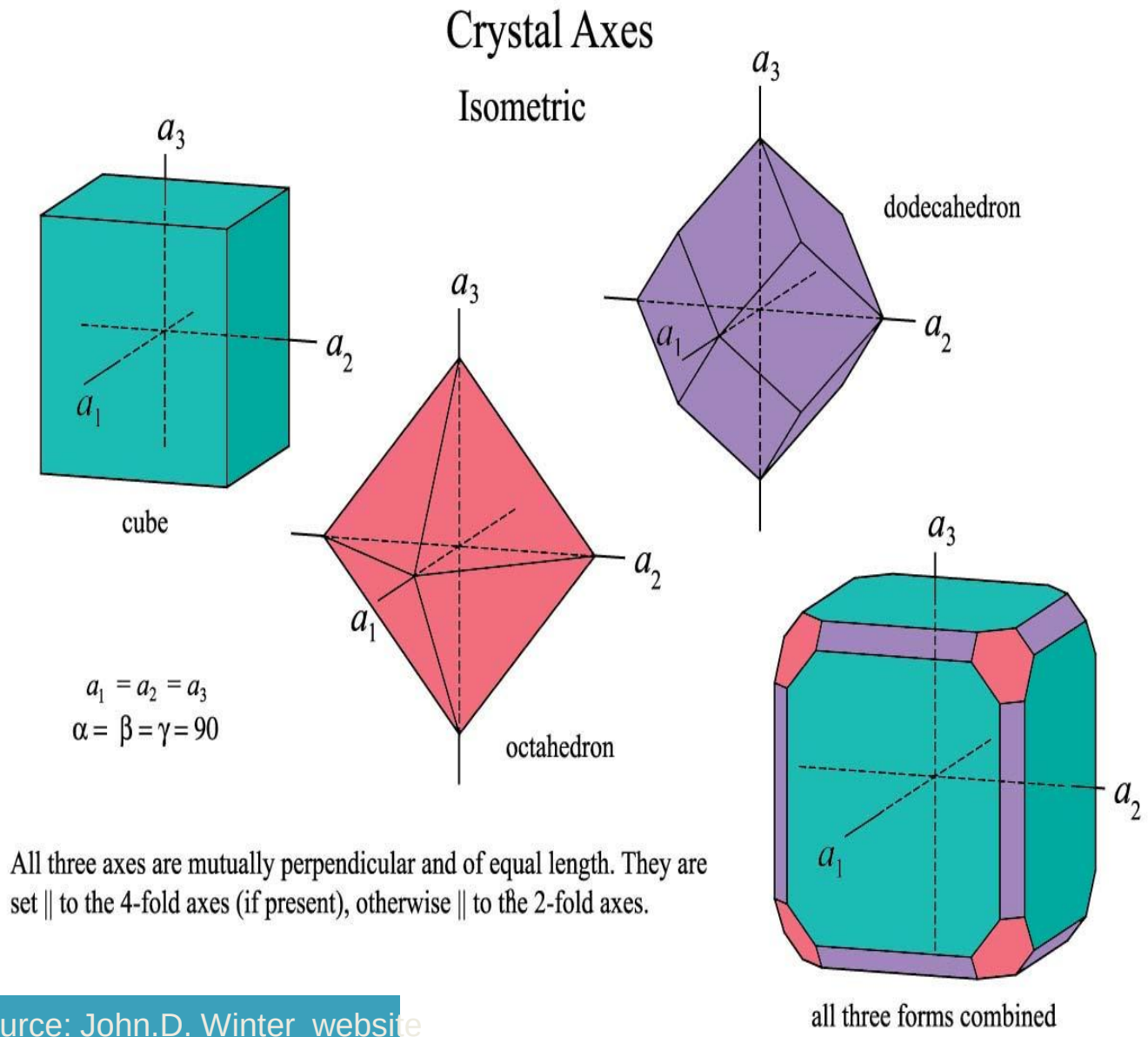


A crystal's unit cell dimensions are defined by six numbers, the lengths of the 3 axes, a, b, and c, and the three interaxial angles, α, β and γ.

Representative Minerals in Crystal Systems

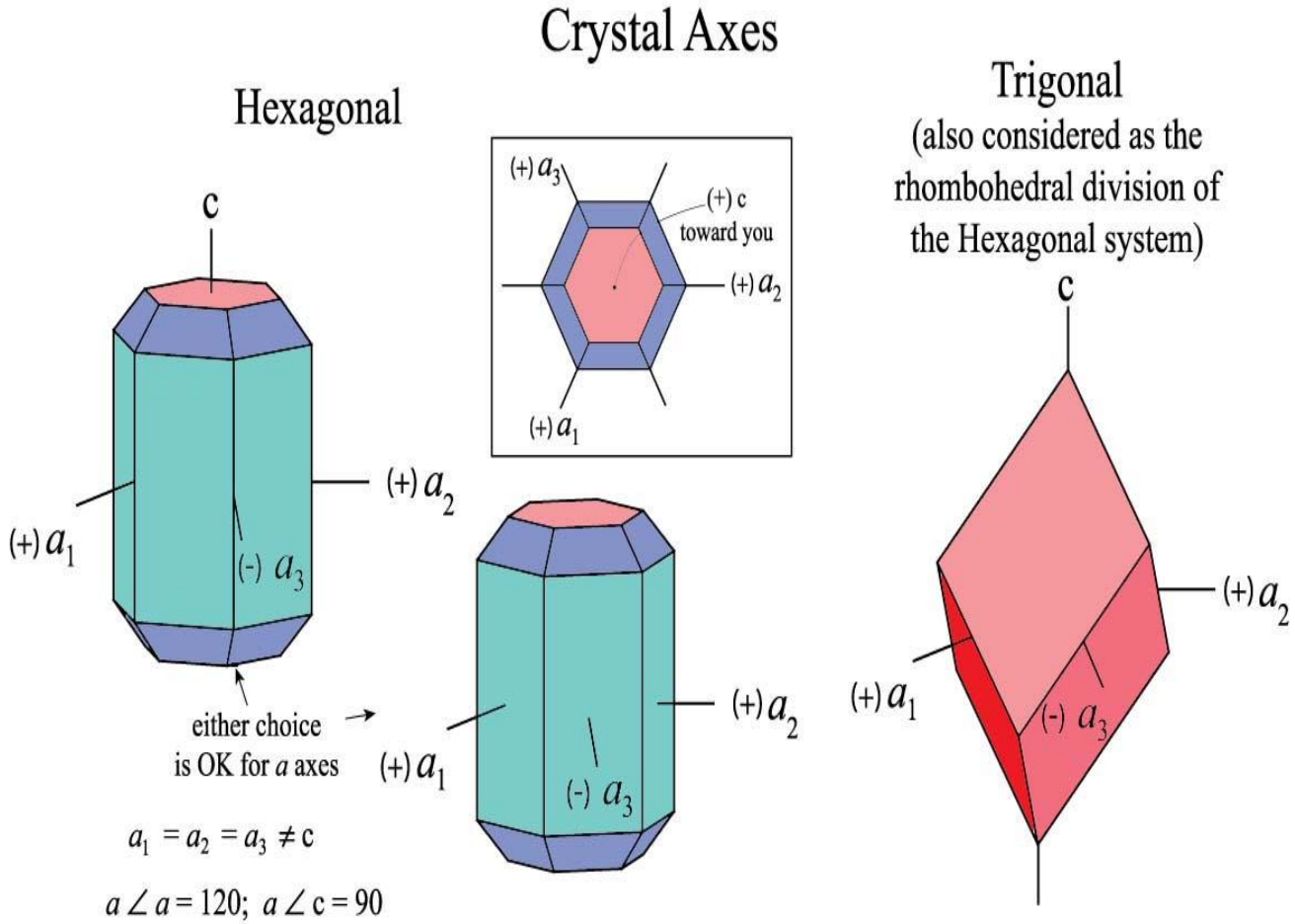


3-D Symmetry



Source: John.D. Winter website

3-D Symmetry

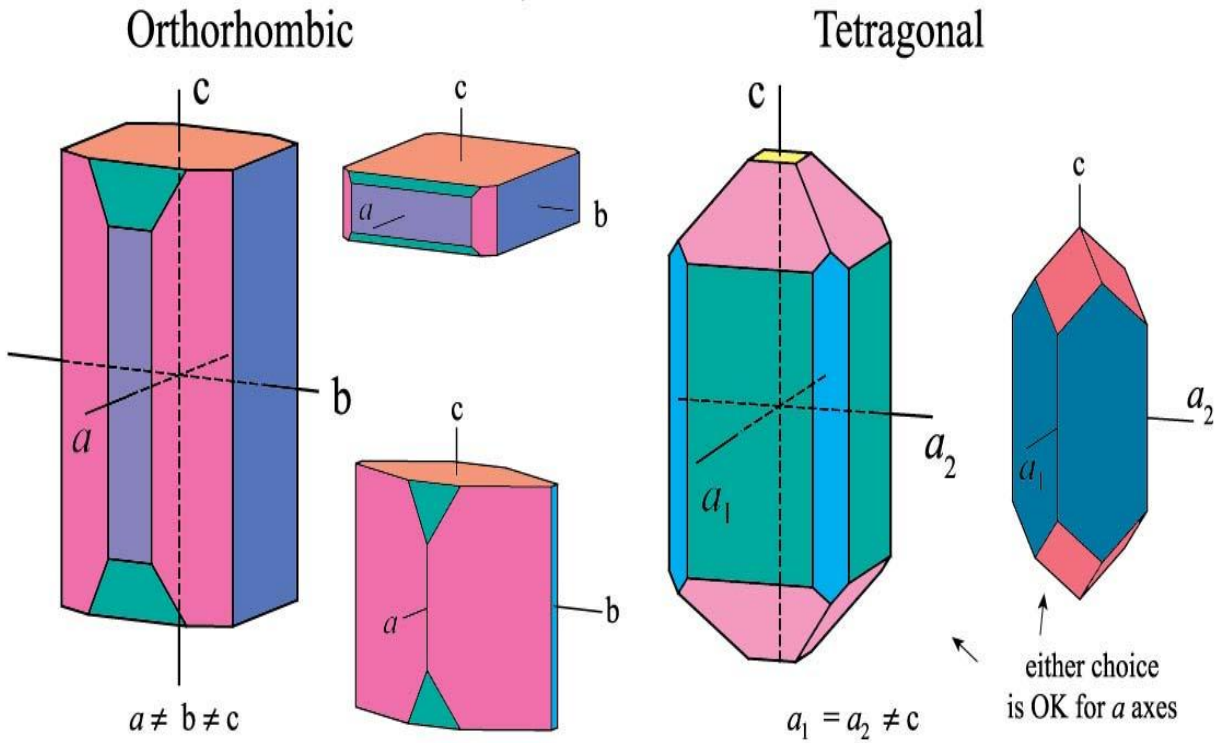


The Hexagonal system (and Trigonal sub-system) typically has four axes, three are of equal length at 120 degrees to one another, and all ⊥ to **c**, which is || to either the 3-fold or 6-fold rotation. The conventional choice of the three ***a*** axes is shown in the inset.

Source: John.D. Winter website

3-D Symmetry

Crystal Axes



$a \neq b \neq c$

$\alpha = \beta = \gamma = 90$

The axes are mutually perpendicular and || to 2-fold axes (conventionally **c** < ***a*** < **b**). When crystals are elongated (as above left) **c** is chosen as the direction of elongation. When crystals are flattened (top insert), **c** is chosen as normal to the predominant plane.

$a_1 = a_2 \neq c$

$\alpha = \beta = \gamma = 90$

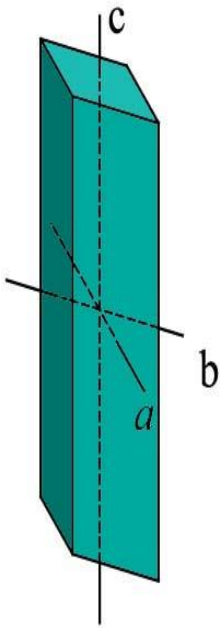
The axes are mutually perpendicular and **c** is chosen || to the 4-fold axis. Due to the 4-fold symmetry, the other two axes are equal. $\boldsymbol{a_1}$ is oriented toward the front and $\boldsymbol{a_2}$ to the right.

Source: John.D. Winter website

Crystallographic Axes & Crystal Systems

Crystal Axes

Triclinic

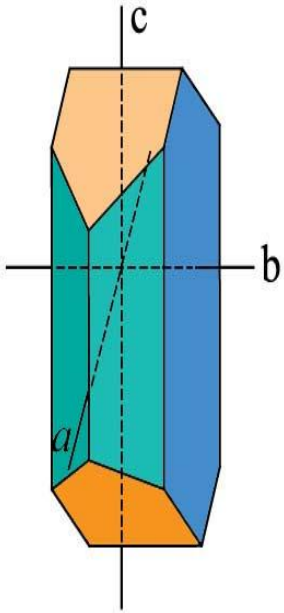


$a \neq b \neq c$

$\alpha \neq \beta \neq \gamma \neq 90$

The axes are chosen as parallel to the principal face intersections. There are no symmetric restrictions to the choice of ***a***, **b**, and **c**, but, by convention, the most pronounced zone is oriented vertically and the zone axis is **c**.

Monoclinic



$a \neq b \neq c$

$\alpha = \gamma = 90, \beta \neq 90$

The axes are chosen as parallel to the principal face intersections. **b** is selected as || to the 2-fold axis or ⊥ to the mirror. The most pronounced zone is oriented vertically and the zone axis is **c**. ***a*** slopes down and forward so that β is typically > 90.

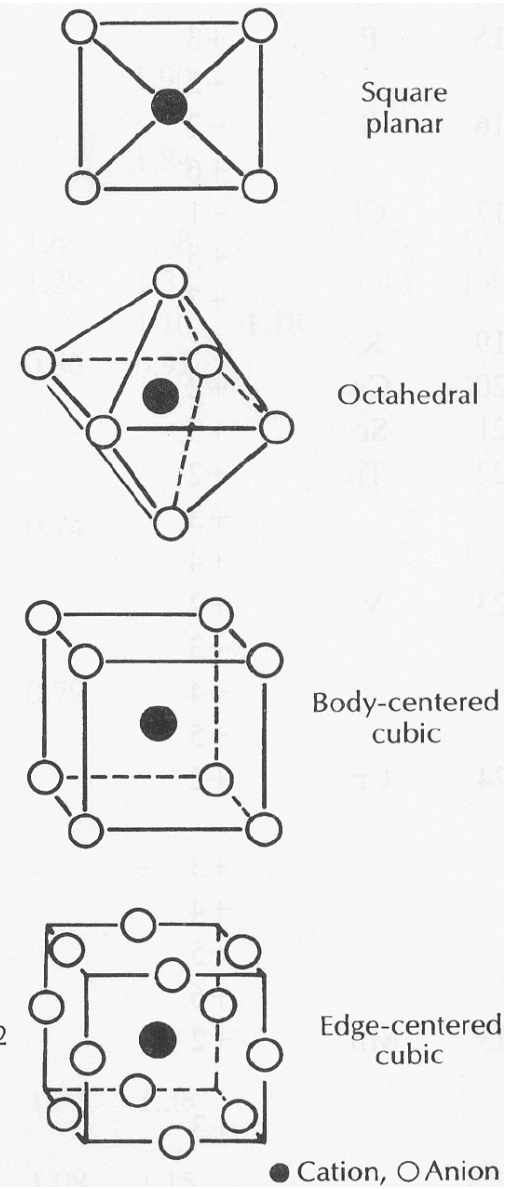
Source: John.D. Winter website

Classification of Minerals

- **Nearly 4000 minerals have been identified on Earth (most of which we don't really worry about).**
- **Rock-forming minerals**
  - **Common minerals that make up most of the rocks of Earth's crust**
  - **Only a few dozen members**
  - **Composed mainly of the 8 elements that make up 98% of the continental crust**

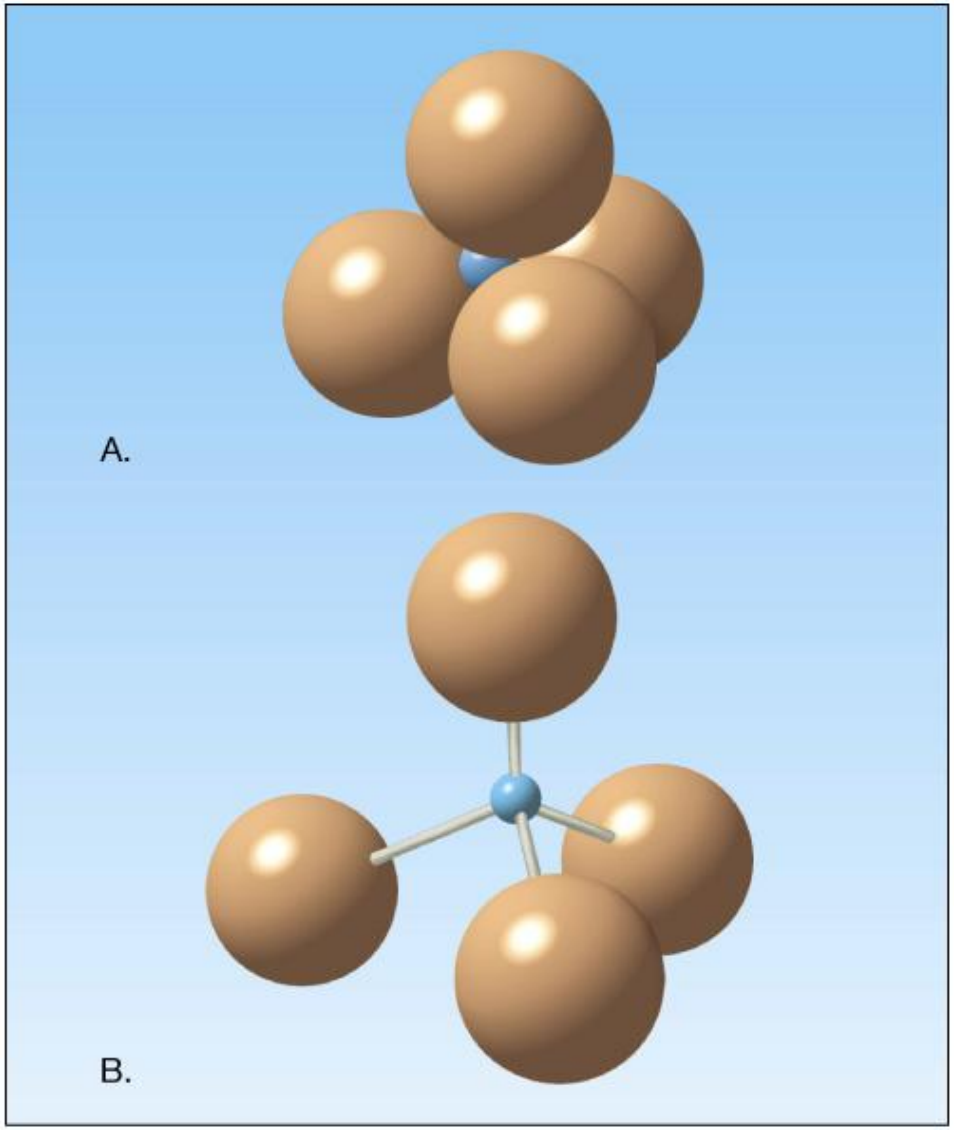
Structure of minerals: Basic considerations are radius ratio and co-ordination number

Radius ratio	Coordination	Arrangement of ions
<0.155	2	Linear
0.155–0.225	3	Trigonal planar
0.225–0.414	4	Tetrahedral
0.414–0.732	4	Square planar
0.414–0.732	6	Octahedral
0.732–1.00	8	Body-centered cubic
>1.00	12	Edge-centered cubic

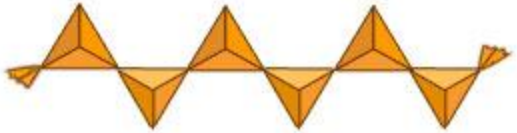
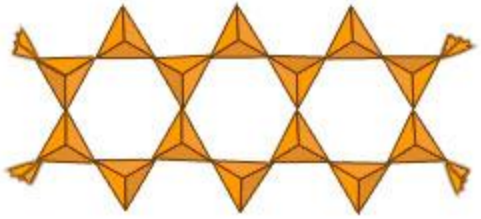
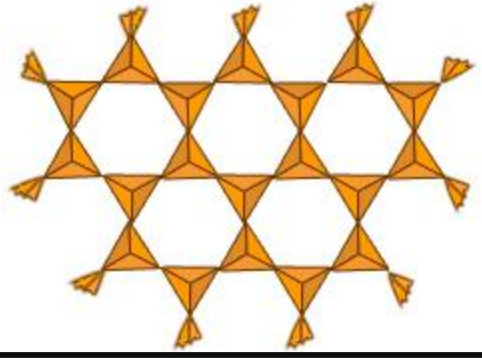


- **Silicates**
  - **Most important mineral group**
    - **Comprise most of the rock-forming minerals**
    - **Very abundant due to large amounts of silicon and oxygen in Earth's crust**
  - **Basic building block is the silicon-oxygen tetrahedron molecule**
    - **Four oxygen ions surrounding a much smaller silicon ion**

Silica Tetrahedra (1 Si, 4 O's)

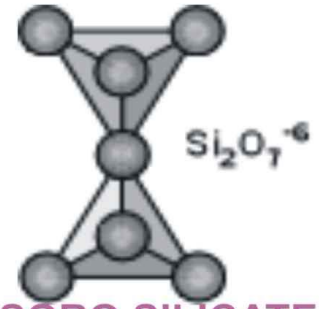


Silica tetrahedron

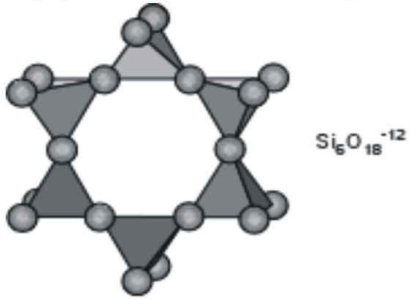
Independent	Olivine		
Single chain	Pyroxene		
Double chain	Amphibole		
Sheet structure	Mica		
Three dimensional network	Quartz, Feldspar		

Linking of SIO4 Tetrahedra

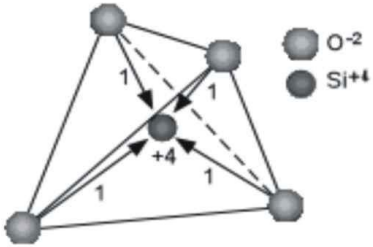
NESO SILICATE
(Olivine, garnet)



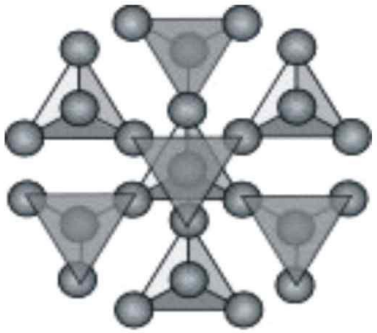
SORO SILICATE
(epidote, melilite)



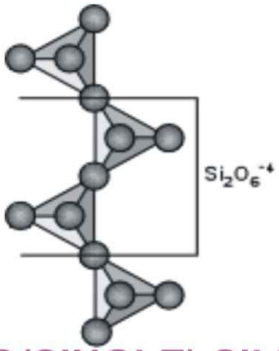
CYCLO/RING SILICATE
(Tourmaline, beryl, cordierite)



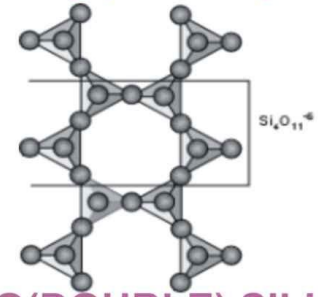
SiO_4^{-4}



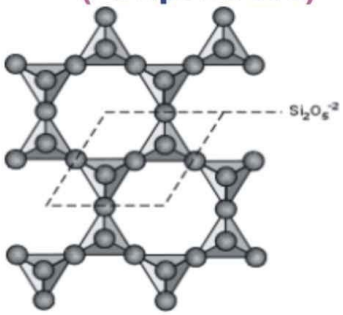
TECTO(FRAME WORK)
SILICATE (Feldspar, quartz)



INO(SINGLE) SILICATE
(pyroxene)

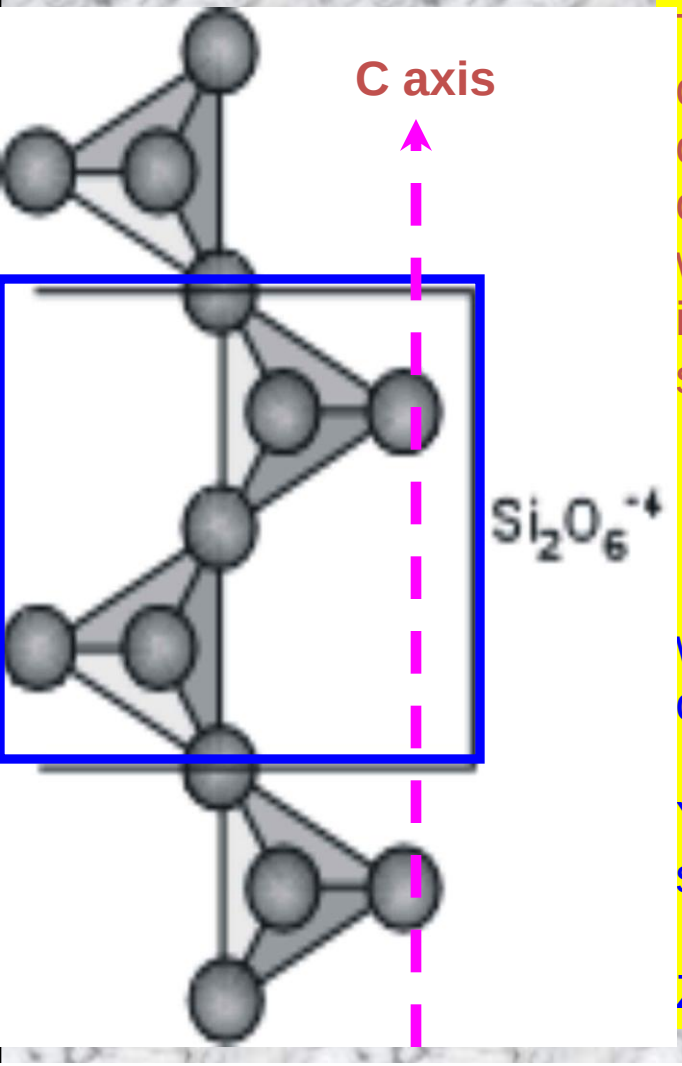


INO(DOUBLE) SILICATE
(Amphibole)



PHYLLO SILICATE (Mica)

SINGLE CHAIN SILICATES : THEIR PHYSICAL AND CHEMICAL PROPERTIES



The single chain silicates have a basic structural unit consisting of linked SiO4 tetrahedra that each share 2 of their oxygens in such a way as to build long chains of SiO4. The basic structural group is thus Si2O6 with an Si:O ratio of 1:3. The most important inosilicates are the pyroxenes. These have a general structural formula of:

XYZ2O6

where X = Na+, Ca+2, Mn+2, Fe+2, or Mg+2 filling octahedral sites called M2

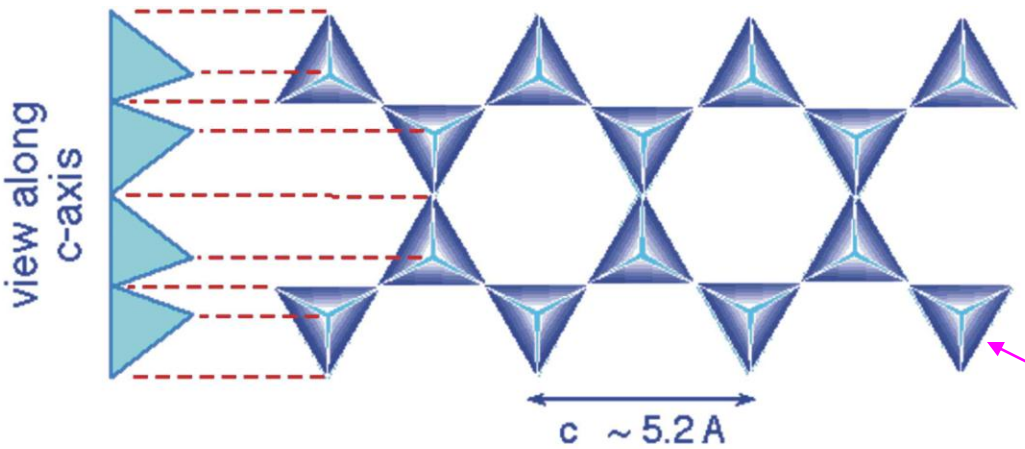
Y = Mn+2, Fe+2, Mg+2 , Al+3, Cr+3, or Ti+4 filling smaller octahedral sites called M1

Z = Si+4 or Al+3 in tetrahedral coordination

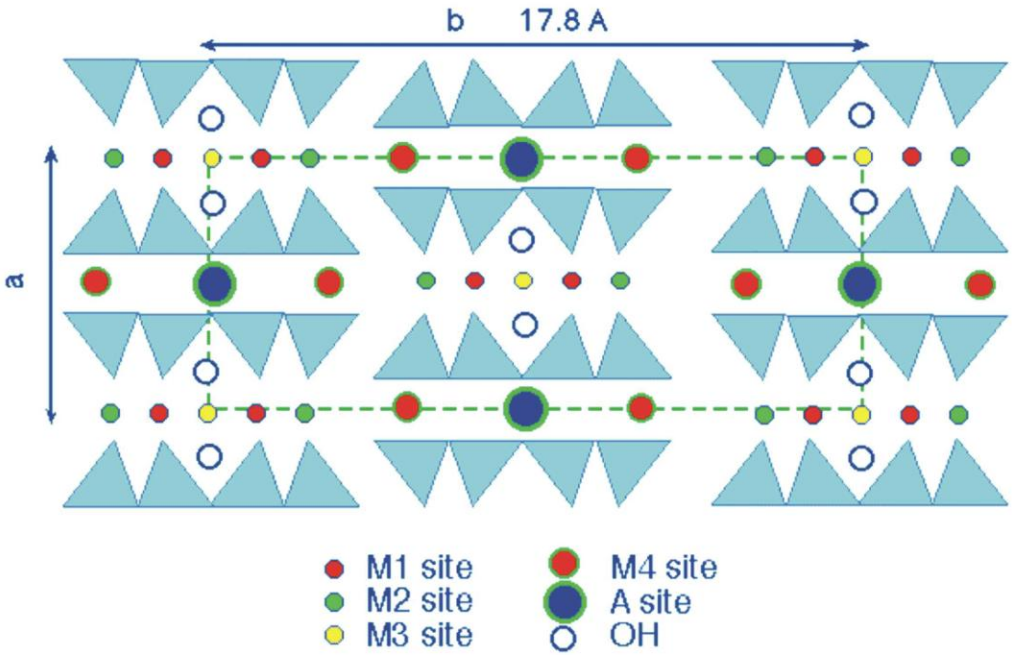
REPEAT UNITS

2- Pyroxene, 3-, 5-, 7- Pyroxenoid Rhodonite (MnSiO3), Wollastonite(CaSiO3)

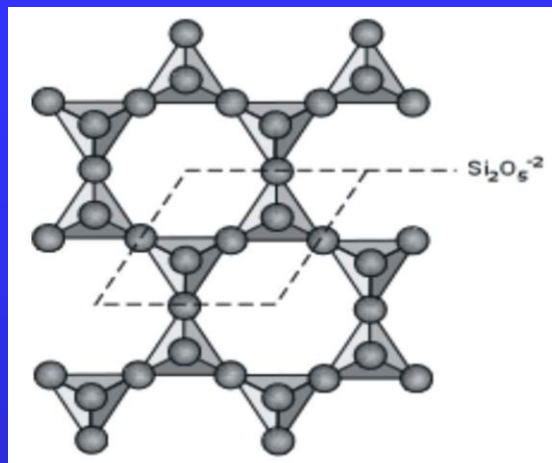
Structure of AMPHIBOLE

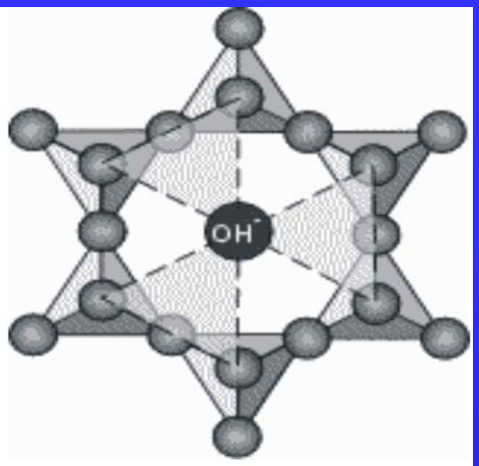


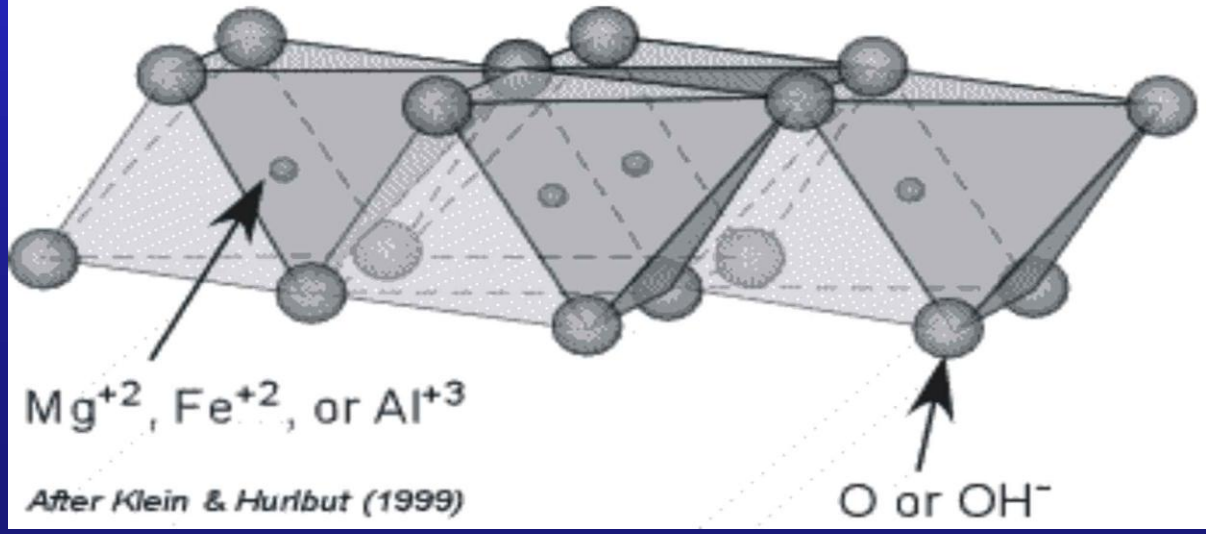
SiO4
tetrahedra



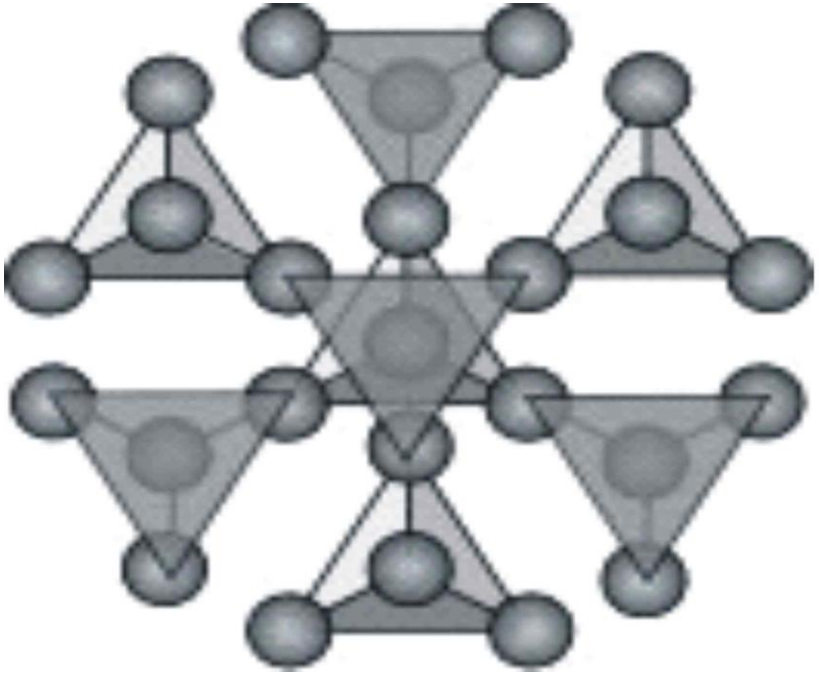
Projection along the C-axis







After Klein & Hurlbut (1999)



All the corners of a SiO4 tetrahedron are linked with the corners of the neighboring tetrahedron resulting in a Si : O = 1:2

Solid solutions

- **Practically all naturally occurring minerals containing Fe-Mg-Mn-Ca or Na-K etc are solid solutions**

  - **Quartz (SiO_2) is not a solid solution**

- **Olivine contains two (there are more) members**
  Forsterite Mg_2SiO_4
  Fayalite Fe_2SiO_4
  Natural olivine is a solid solution of FO-Fa

Ionic Substitution

Substitution is possible if difference in radii is < 15%

Substitution is possible between ions of same charge, or else through coupled substitution maintaining charge balance

If two ions compete for the same site, the one with higher ionic potential is preferred

Substitution can be limited if there is a large difference in electronegativity or difference in bonding character between two ions of same charge and similar size.

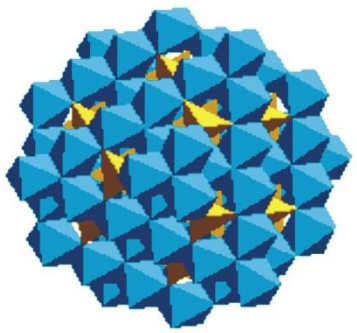
Ionic Substitution leads to formation of solid solution

Some interesting details about Olivine

- **Nesosilicate: Isolated SiO_4 Tetrahedra**
- **Most common mineral in the Earth's mantle**
- **$(Mg,Fe)_2SiO_4$**
- **Mg, Fe form MO_6 Octahedra that link Tetrahedra**
- **Olivine is Orthorhombic near surface conditions called Alpha olivine**
- **At higher P-T, Alpha olivine changes to Beta olivine which contains sites that contain (OH)**
- **At still higher P-T, changes to gamma olivine which is isometric and can contain OH**
- **At still higher P-T changes to perovskite structure**

High Pressure Polymorph of Olivine

RINGWOODITE

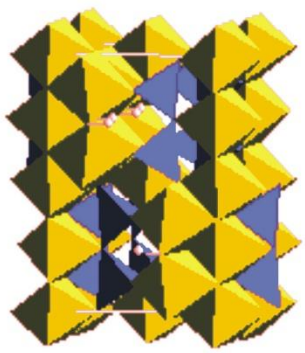


(Fe,Mg)2SiO4 in the spinel (Cubic) structure is known as ringwoodite (γ). Here, Si atoms occupy the tetrahedral sites (yellow) and Fe and Mg atoms occupy the *edge-sharing* octahedral sites (blue).

SPINEL (AB2O4)

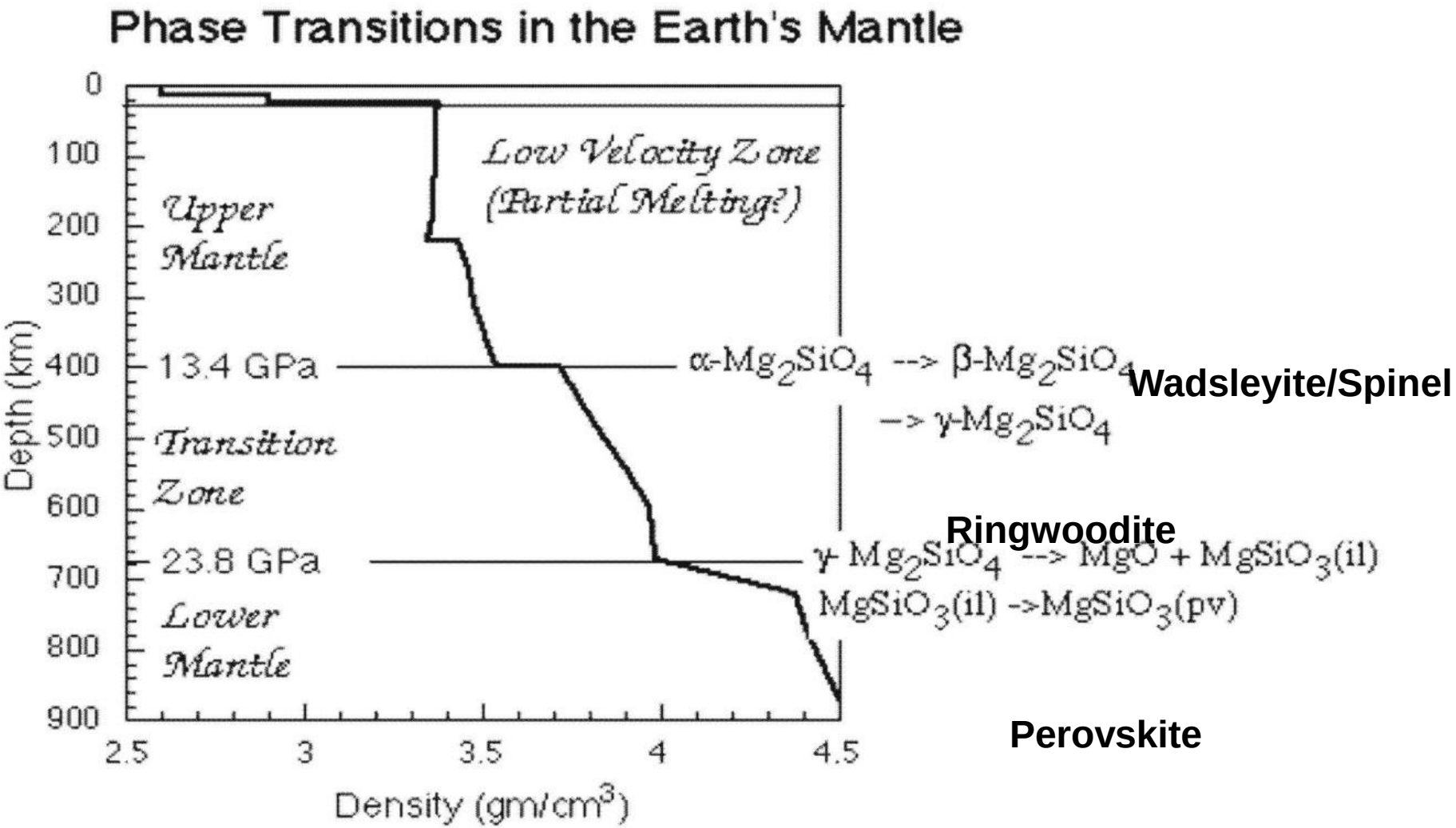
SiMg2O4

WADSLEYITE

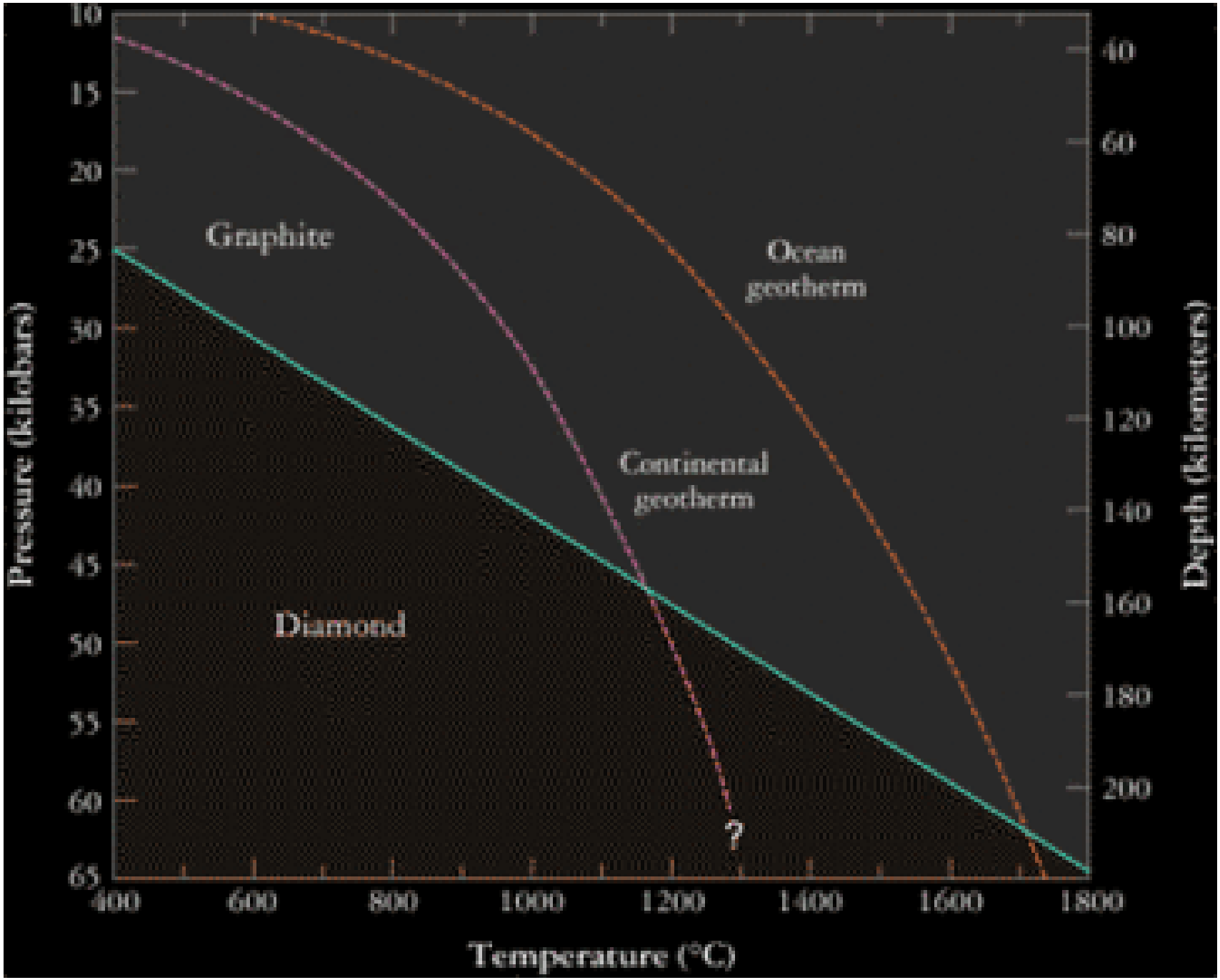


OH in the structure

Intermediate between the olivine and spinel (ringwoodite) structure is a phase known as wadsleyite (β (Mg,Fe)2SiO4) The wadsleyite (or beta-spinel) structure is currently of great interest because it is being invoked as a reservior of water (as chemically bound OH) in the Earth's mantle.

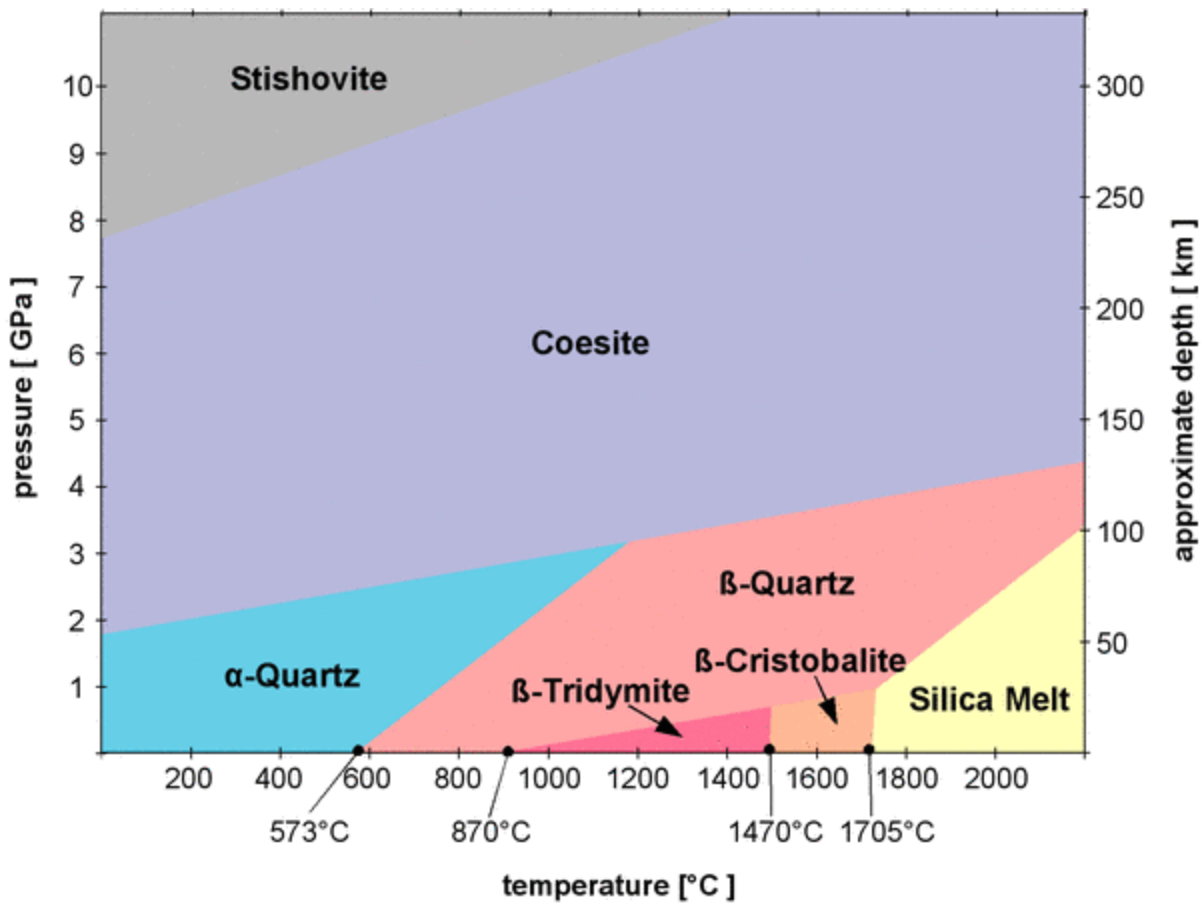


Alpha to Beta at 410 Km, Beta to Gamma at 520 Km, Gamma to Perovskite at 660 Km


Graphite
Diamond
Ocean geotherm
Continental geotherm
?
Pressure (kilobars)
Depth (kilometers)
Temperature (°C)
10
15
20
25
30
35
40
45
50
55
60
65
40
60
80
100
120
140
160
180
200
400
600
800
1000
1200
1400
1600
1800

SiO_2

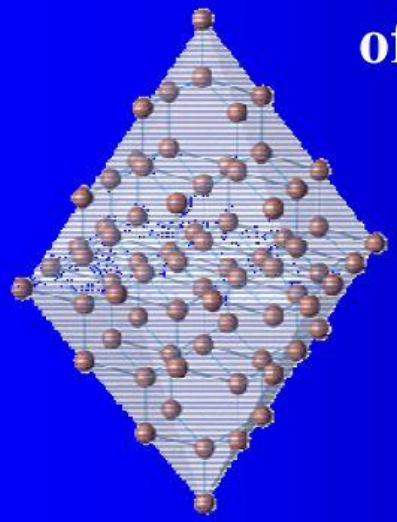
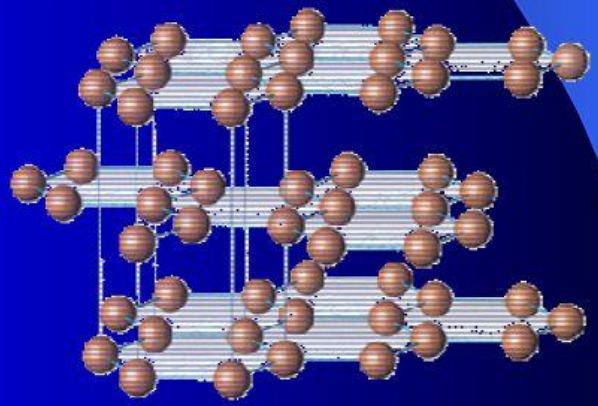
Polymorphism



Structure of Minerals

Diamond

Graphite

Polymorphs

of Carbon